

# Math 289, Fall 2026

## Lecture 1

**Problem 1.** Four people  $A$ ,  $B$ ,  $C$  and  $D$  need to get across a river. The only way to cross the river is by an old bridge, which holds at most 2 people at a time. Being dark, they can't cross the bridge without a torch, of which they only have one. So each pair can only walk at the speed of the slower person. They need to get all of them across to the other side as quickly as possible.  $A$  is the slowest and takes 10 minutes to cross;  $B$  takes 5 minutes;  $C$  takes 2 minutes; and  $D$  takes 1 minute.

What is the minimum time to get all of them across to the other side?

**Problem 2.** You and your colleagues know that your boss  $A$ 's birthday is one of the following 10 dates:

|       |       |       |
|-------|-------|-------|
| Mar 4 | Mar 5 | Mar 8 |
| Jun 4 | Jun 7 |       |
| Sep 1 | Sep 5 |       |
| Dec 1 | Dec 2 | Dec 8 |

$A$  told you only the month of his birthday, and told your colleague  $C$  only the day. After that, you first said: "I don't know  $A$ 's birthday;  $C$  doesn't know it either." After hearing what you said,  $C$  replied: "I didn't know  $A$ 's birthday, but now I know it." You smiled and said: "Now I know it, too." After looking at the 10 dates and hearing your comments, your administrative assistant wrote down  $A$ 's birthday without asking any questions.

So what did the assistant write?

**Problem 3.** How many trailing zeros are there in  $100!$ ?

**Problem 4.** A grasshopper starts at the origin in the coordinate plane and makes a sequence of hops. Each hop has length 5, and after each hop the grasshopper is at a point whose coordinates are both integers; thus, there are 12 possible locations for the grasshopper after the first hop.

What is the smallest number of hops needed for the grasshopper to reach the point  $(2021, 2021)$ ?

**Problem 5.** There are five houses of different colors next to each other. In each house lives a man. Each man has a unique nationality, an exclusive favorite drink, a distinct favorite brand of cigarettes and keeps specific pets. Use all the clues below to fill the grid and answer the question: "Who owns the fish?"

1. The Brit lives in the red house.
2. The Swede keeps dogs as pets.
3. The Dane drinks tea.
4. The green house is immediately to the left of the white house.
5. The owner of the green house drinks coffee.
6. The person who smokes Pall Mall rears birds.

7. The owner of the yellow house smokes Dunhill.
8. The man living in the center house drinks milk.
9. The Norwegian lives in the first house.
10. The man who smokes Blends lives next to the one who keeps cats.
11. The man who keeps horses lives next to the man who smokes Dunhill.
12. The owner who smokes Blue Master drinks beer.
13. The German smokes Prince.
14. The Norwegian lives next to the blue house.
15. The man who smokes Blends has a neighbor who drinks water.